
Software Requirements Specification

for

Mentor Caring System

Version 1.0 approved

Prepared by Ma Xinyao, Guan Xinling, Hou Shuoran, Peng
Yancheng, Tang Jianxu, Ouyang Hao

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Revision History

Version	Date	Author	Description
1.0	2026.03.12	B09	Initial approved version

1. Introduction

1.1 Purpose

This document provides a detailed description of the requirements for the **Mentor Caring System (MCS)**. The purpose of this Software Requirements Specification (SRS) is to describe the systems functions, external interfaces, constraints, and overall behavior.

This document is intended for project stakeholders including developers, project managers, instructors, and future maintainers of the system. It serves as a reference for system design, implementation, and testing.

1.2 Document Conventions

This document follows standard Software Requirements Specification conventions. Section titles and numbering follow the structure defined in the SRS template. Important terms and system components are written with consistent naming throughout the document.

Diagrams such as the context model and use case diagrams are included to visually represent system structure and user interactions.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following readers:

1. **Developers**, who will design and implement the Mentor Caring System;
2. **Project team members**, who need a shared understanding of the system scope and requirements;
3. **Instructors and evaluators**, who will review whether the requirements are complete, clear, and consistent with the project description;
4. **Future maintainers**, who may extend or revise the system in later stages.

Readers who want a general understanding of the system should begin with Section 2: Overall Description.

Readers who are interested in system integration and external dependencies should focus on Section 4: External Interface Requirements.

Readers who want to understand quality constraints and operational expectations should read Section 5: Other Nonfunctional Requirements.

1.4 Project Scope

The **Mentor Caring System (MCS)** is a web-based information system designed to support the **BNBU Mentor Caring Programme**, which aims to help undergraduate students adapt to university life more effectively through structured mentoring and follow-up activities. The system is intended to provide a centralized platform for managing mentoring-related information, communication, and records for the different participants involved in the programme.

The system supports multiple types of users, including **Administrator**, **Faculty Consultant**, Mentor, Student, MCP Coordinator, and optionally **Supporting Staff**. It is intended to improve the efficiency and consistency of mentoring management by digitalizing major mentoring-related activities.

The main goals of the system are to:

- allocate mentors;
- report and maintain interview records;
- support appointment arrangement between mentors and students;
- generate summary reports;
- track relevant user and mentoring information.

To achieve these goals, the system supports key mentoring management activities carried out by different user roles, including administrators, faculty consultants, mentors, students, and MCP coordinators. It is expected to improve the efficiency, consistency, and traceability of mentoring work by integrating mentor allocation, record management, appointment support, information search, and communication functions into one system. The system also helps preserve historical mentoring records and supports the import of organizational and mentoring data, which is important for maintaining continuity when mentoring assignments or group membership change.

From the university's perspective, MCS supports the operational objectives of the BNBU Mentor Caring Programme by enabling more systematic management of mentoring activities, improving communication among participants, and making student support information easier to maintain and follow up. In this way, the system contributes to a more coordinated and scalable mentoring process within BNBU.

1.5 References

Design and Implementation of a Mentor Caring System, Project Long Description, Version 1.0, 2025–2026 Semester 2.

2. Overall Description

2.1 Product Perspective

The Mentor Caring System is a new information system developed for the BNBU Mentor Caring Programme. It is designed to support mentor caring activities for undergraduate

students, including mentor allocation, record management, appointment arrangement, and communication.

The system is not intended to replace an existing product. It operates within the university environment and relies on existing institutional resources, especially the university account database for login and user identification, as well as related student and staff information sources.

The system also interacts with external supporting components, such as Excel files for data import and the university email or communication system for user notification. The context model in this section shows the system boundary and its main external interfaces.

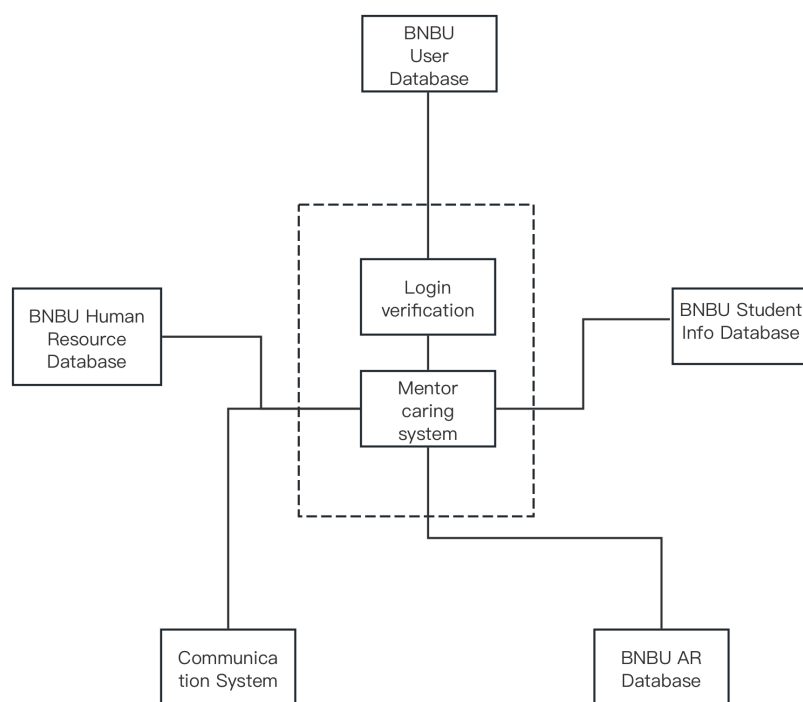


Figure 1: Context Model of the Mentor Caring System

2.2 Product Features

The Mentor Caring System supports the main activities of the BNBU Mentor Caring Programme. Its major features include role-based user access, mentor-group management, information search, interview record management, appointment arrangement, communication between users, special case forwarding, and record export. These functions help different users complete mentor caring tasks efficiently through one integrated system.

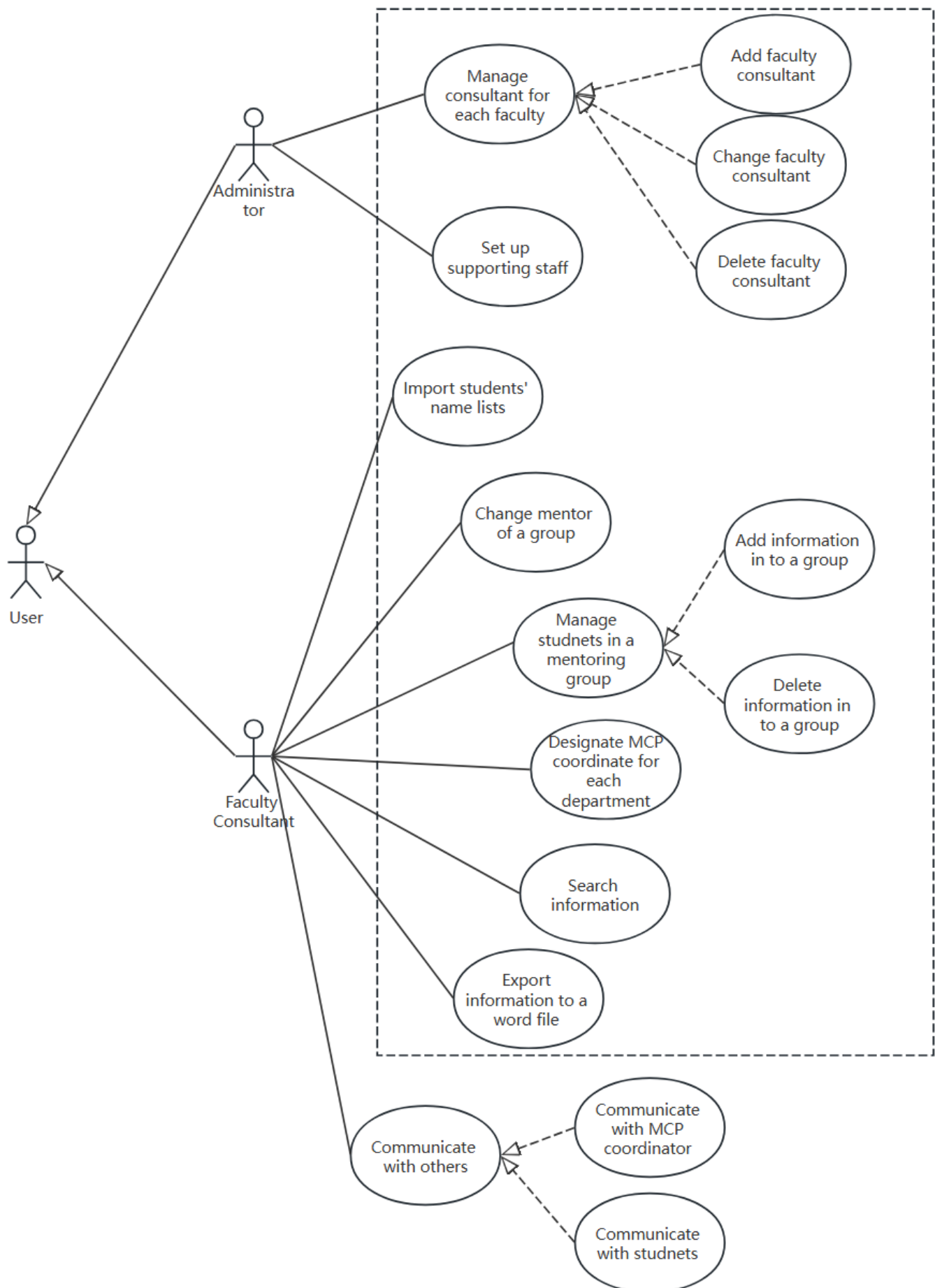


Figure 2: Use Case Diagram of the Mentor Caring System

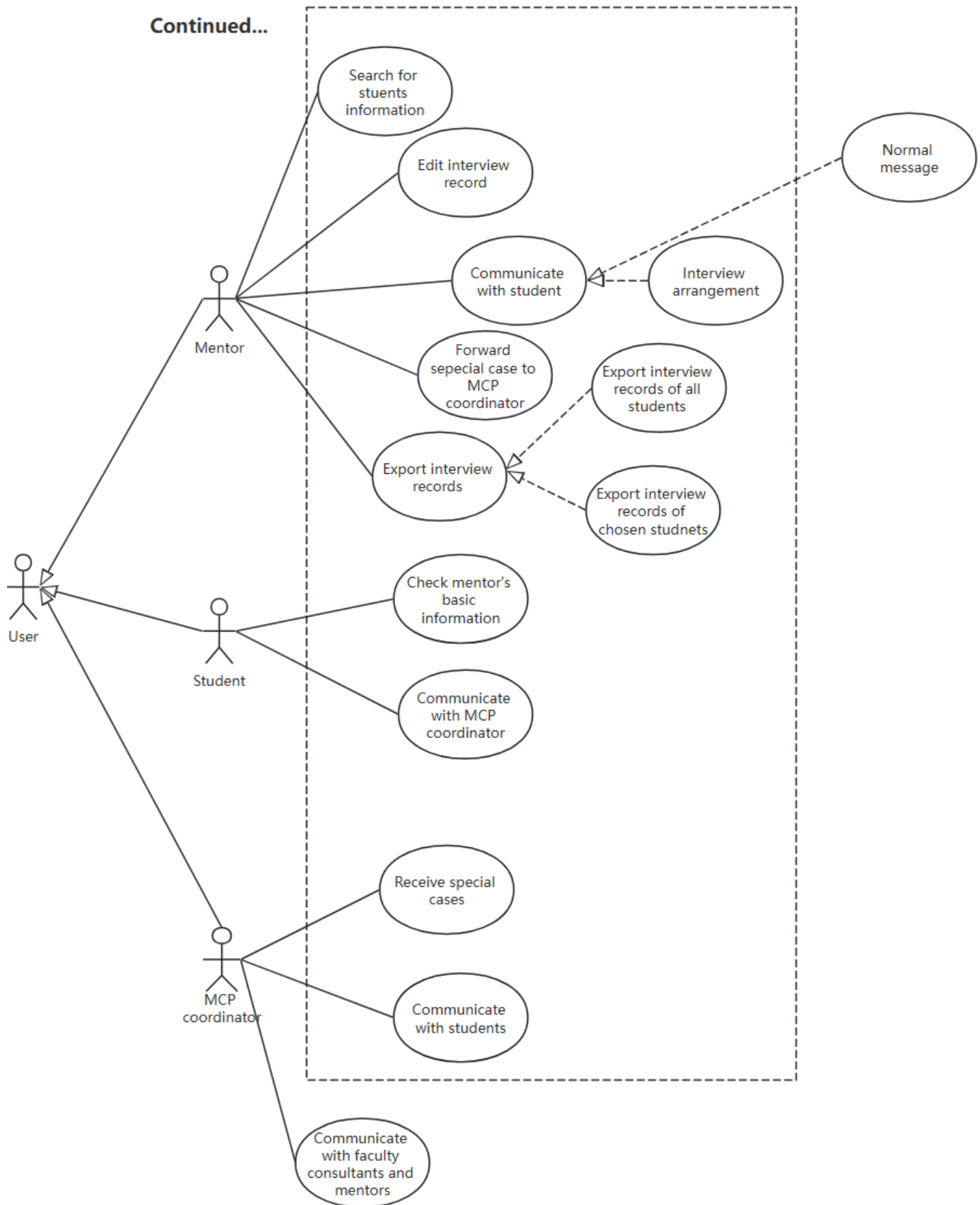


Figure 2. cont: Use Case Diagram of the Mentor Caring System

2.2.1 Scenario for “Search Mentor Info”

Plan A – Basic Scenario

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.
6. The system displays the mentor’s name associated with the group.
7. The system displays the record of one student in that group.

Plan B – Alternative Scenario: Group Not Found

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database.
6. The system cannot find the group information.
7. The system displays a message indicating that the Group ID is invalid and the use case ends.

Plan C – Alternative Scenario: No Student Record

1. The user logs in (see the Login scenario).
2. The user activates the “Search Mentor Info” function.
3. The system displays the input box for the Group ID.
4. The user enters the Group ID.
5. The system searches the database for the group information.
6. The system finds the mentor information associated with the group.
7. The system cannot find any student record in that group.
8. The system displays the mentor’s name and a message indicating that no student record is available.

2.3 User Classes and Characteristics

The main user classes of the Mentor Caring System are administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.

- Administrator

The administrator is a privileged user responsible for overall system setup and management. Usually there is only one administrator in the university. This user designates and manages faculty consultants and supporting staff. The administrator has the highest management authority but is expected to use the system less frequently than faculty consultants, mentors, and students.

- Faculty Consultant

Faculty consultants are important management users at the faculty level. They are responsible for maintaining mentor allocation, managing mentoring groups, assigning MCP coordinators, searching information, and exporting records. They are frequent users, especially at the beginning of each academic year and whenever group allocation changes are needed.

- **Mentor**

Mentors are core operational users of the system. They mainly use the system to view students in their own groups, maintain interview records, communicate with students, arrange interviews, and handle special cases. They are frequent users and generally have moderate computer literacy. Their access rights are limited to their own groups.

- **Student**

Students are end users of the system. Any current undergraduate student can log in to the system, but only students assigned to a mentoring group can continue to use its main functions. Students mainly use the system to check their mentors basic information and communicate with mentors. This user class is large in number and requires a simple and easy-to-use interface.

- **MCP Coordinator**

MCP coordinators are department-level users designated by faculty consultants. They are responsible for searching mentor and student information, handling special cases, and communicating with related users. In some cases, an MCP coordinator may also serve as a mentor. This user class has moderate to high access privileges.

- **Supporting Staff**

Supporting staff are optional users set up by the administrator. They mainly view student log information and may respond to user feedback about system usage. Since this function is optional, this user class is less important than the main operational user classes, and its usage frequency is expected to be relatively low.

2.4 Operating Environment

- The system depends on an existing school database that stores staff and student accounts, passwords, names, and roles.
- The system must support the import of Excel files for organization structure, mentor allocation, and MCP coordinator setup.
- The system must also support exporting information and interview records to Word files.
- The system must support the university's email service to send email notification.

2.5 Design and Implementation Constraints

- The system shall use existing university accounts for user login rather than creating an independent account system.
- Import file format constraints: the system should strictly import file of Excel for setting faculties, departments, majors, student-mentor allocations, and MCP coor-

dinators.

- Export file format constraints: the system should strictly export selected information and interview records to Word files.
- The system shall preserve historical student information even when mentors are changed or students are removed from groups.
- The system design shall be compatible with the existing school database and related university data sources.
- The system shall enforce role-based access restrictions for people with specific role could only access to specific functions and data scopes.

2.6 User Documentation

- An online user guide should be provided for all major user classes, including administrator, faculty consultant, mentor, student, MCP coordinator, and supporting staff.
- The user documentation should include instructions for login using university accounts and basic navigation of the system.
- The user documentation should provide step-by-step guidance for faculty consultants on importing Excel files, including the formats for organization structure setup, student-mentor allocation, and MCP coordinator setup.
- The user documentation should explain how to search for student information, mentor information, and log information for the relevant user roles.
- The user documentation should explain how mentors create, modify, and delete interview records, and how they arrange interview appointments with students.
- The user documentation should explain the internal messaging functions, including sending messages, receiving email notifications, logging in to read messages, and replying to messages.
- The user documentation should include instructions for exporting information and interview records to Word files.

2.7 Assumptions and Dependencies

- It is assumed that a university database containing valid staff and student accounts, passwords, names, and user roles is already available for authentication and identity management.
- It is assumed that faculty consultants will provide correct and properly formatted Excel files for setting up.
- It is assumed that student email addresses can be derived or obtained from student IDs.

3. System Features

3.1 System Feature 1

3.1.1 Description and Priority

3.1.2 Stimulus/Response Sequences

3.1.3 Functional Requirements

REQ-1

REQ-2

3.2 System Feature 2 (and so on)

4. External Interface Requirements

4.1 User Interfaces

4.2 Hardware Interfaces

The **Mentor Caring System (MCS)** does not require direct control over specialized hardware. It relies on the underlying hardware of client devices and hosting servers to facilitate user interactions and data processing.

4.2.1 Supported Client Devices

Logical Characteristics: The system supports any device capable of running a modern web browser.

Physical Characteristics: Supported devices include desktop computers, laptops, tablets, and smartphones. The physical interfaces involve standard human-machine interface components: keyboards, mice, trackpads, and touchscreens for input, and standard displays for output.

Hardware Communication: Client devices connect to the server via standard physical network interface cards (NICs) supporting Ethernet (IEEE 802.3) or Wi-Fi (IEEE 802.11).

4.2.2 Server Hardware Interfaces

The **Mentor Caring System (MCS)** will be deployed on BNBU's server hardware. The server hardware must provide sufficient memory and CPU resources to handle concurrent database queries and file I/O operations, including parsing Excel files and generating Word documents.

4.2.3 Data and Control Interactions

User inputs are logically captured by the client browser and converted into HTTP/HTTPS requests sent to the server. The server hardware processes these requests and returns HTML/JSON data, which the client hardware renders onto the screen.

4.3 Software Interfaces

The **Mentor Caring System (MCS)** must communicate with several distinct software components to fulfill its functional requirements.

4.3.1 School Databases

Component & Version: BNBU User Authentication Database (version TBD, maintained by university IT).

Data Coming In: The MCS retrieves user profile data, including staff and student accounts, passwords, names, and roles (staff or student).

Data Going Out: The MCS sends authentication validation requests, including user account and password input, to the database.

Purpose & Services: To authenticate users upon login and establish their roles (**Administrator, Faculty Consultant, Mentor, Student, MCP Coordinator**) without duplicating user accounts.

Nature of Communications: Communication will be handled via secure backend API calls.

4.3.2 Email System

Component & Version: BNBU SMTP Mail Server.

Data Going Out: Email trigger messages containing recipient addresses, standard subject lines, and static notification body text.

Purpose & Services: To notify users when they receive a normal message, an interview appointment, or a forwarded special case.

4.3.3 Document Processing Libraries

Component & Version: Standard open-source file parsing libraries (e.g., Apache POI for Java or openpyxl/python-docx for Python, version 4.0+).

Data Coming In (Excel): .xls or .xlsx files containing organizational architecture (faculties, departments, majors), student-mentor allocation lists, and MCP coordinator information, as specified in Appendices A, B, and C.

Data Going Out (Word): .doc or .docx files containing exported interview records and summary reports, as specified in Appendix D.

Implementation Constraint: The file reading and writing mechanism requires local temporary storage space on the server operating system before transferring the stream to the client browser.

4.3.4 Operating Systems

Server Operating Systems: The backend application and database will run on a standard server operating system, such as Linux Ubuntu 22.04 LTS.

Client Operating Systems: Windows, macOS, iOS, and Android, operating independently from the server via web browsers.

4.4 Communications Interfaces

The **Mentor Caring System (MCS)** requires robust communication interfaces to ensure that interactions between users and the server are secure, fast, and reliable.

4.4.1 Web Browser and Network Protocols

Standards Used: All communication between the client web browser and the MCS server will utilize HTTP/1.1 or HTTP/2 over TCP/IP.

Electronic Forms: Users will submit data (e.g., sending messages, adding or deleting students, and making appointments) using standard HTML5 electronic forms.

4.4.2 Communication Security and Encryption

HTTPS: To prevent data loss and malicious attacks, all HTTP traffic must be encrypted using TLS 1.2 or higher (HTTPS). This is crucial to protect student privacy and login credentials.

Database Security: Communication between the MCS backend and the BNBU school database must be restricted to the university's secure internal network and protected by firewall rules.

4.4.3 Email Communication Standards

Protocol: SMTP (Simple Mail Transfer Protocol) will be used to push email notifications to the BNBU email server.

4.4.4 Data Transfer Rates and Synchronization

Synchronization Mechanisms: Email notifications will be sent asynchronously using a background message queue to prevent the system from freezing while waiting for the email server to respond.

Data Transfer Rates: File uploads and downloads must support a minimum transfer rate to avoid session timeouts. The system will define a timeout standard of 30 seconds for standard queries.

Message Formatting: API communications between the frontend user interface and backend server will utilize JSON formatting for lightweight and efficient data exchange.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Mentor Caring System shall provide acceptable performance for administrators, faculty consultants, mentors, MCP coordinators, students, and supporting staff during normal system operation. The system shall support login, search, messaging, appointment management, Excel import, Word export, and log recording while maintaining reasonable response times. The system shall also remain usable during yearly mentor allocation updates while preserving historical records.

5.1.1 Response Time

- User login authentication shall complete within 3 seconds under normal network conditions.
- Search queries for student, mentor, or log information shall return results within 3 seconds.
- Student records or mentoring group member lists shall display within 2 seconds after selection.
- Sending a message shall complete within 2 seconds, excluding external email delivery delay.
- Appointment confirmation shall complete within 2 seconds after submission.

5.1.2 Appointment Processing

- The system shall support appointment booking based on mentor-defined available time slots.
- Each appointment slot shall represent 30 minutes.
- Available time slots shall be displayed to students without noticeable delay.
- Once an appointment is confirmed, the system shall immediately update and display the appointment status to relevant users.

5.1.3 Import and Export Efficiency

- Import of valid Excel files for mentor allocation or MCP coordinator setup shall complete within 30 seconds for a normal dataset.
- Export of student or interview records to a Word file shall complete within 30 seconds.
- Faculty consultants shall be able to export records filtered by academic year, department or major, mentor name, and student name.

5.1.4 Concurrent Usage

- The system shall support at least 100 concurrent active users without major degradation of performance.

- User operation logs shall be recorded in the background without interrupting foreground operations.
- Normal user operations shall remain responsive while yearly mentor allocation data are being imported.

5.2 Safety Requirements

Although the Mentor Caring System is not a physical safety-critical system, incorrect operations may lead to loss of student records, incorrect mentor assignment, or missed interviews. Therefore, the system shall include safeguards to prevent data loss, inconsistent appointments, and unauthorized destructive actions while preserving historical records.

5.2.1 Data Preservation

- Historical student information shall be preserved when annual mentor allocation data are imported.
- Student information shall remain preserved even if the student is removed from a mentoring group.
- Authorized mentors shall be able to access relevant historical records after reassignment.
- The system shall prevent accidental permanent deletion of valid student records.

5.2.2 Operational Controls

- Confirmation shall be required before critical operations such as deleting a faculty consultant, removing a student from a group, changing mentor assignments, or forwarding special cases.
- Imported Excel files shall be validated before updating system data.
- Invalid files or incorrect formats shall be rejected with clear error messages.
- Existing valid information shall not be overwritten when duplicated data appear in imported files.

5.2.3 Appointment and Communication Safety

- The system shall prevent double booking of appointment time slots for the same mentor.
- Confirmed appointment time and venue shall be visible to relevant users.
- Email notifications shall only notify users that a message exists and shall not replace reading the message inside the system.

5.2.4 Warning and Recovery

- If a student logs in but is not assigned to any mentoring group, the system shall display a warning and require the student to exit.
- Operation logs shall be maintained to support tracing of user actions.

- If import, export, or record updates fail, the system shall notify the user and prevent corrupted data.

5.3 Security Requirements

The Mentor Caring System manages student records, interview records, communication data, appointment information, and activity logs. The system shall therefore provide authenticated access, role-based authorization, privacy protection, and auditability.

5.3.1 Authentication

- All users shall be authenticated using valid university accounts and passwords from the school database.
- Successful authentication shall be required before accessing protected functions.
- Users shall log in to read messages even if email notifications are received.

5.3.2 Authorization

- The system shall enforce role-based access control so users can only access functions and data permitted to their role.
- Faculty consultants shall manage only information related to their authorized faculty.
- Mentors shall access only students in their mentoring groups during their mentoring period.
- Students shall access only their own mentoring information.
- MCP coordinators and supporting staff shall access only the information required for their assigned responsibilities.

5.3.3 Privacy Protection

- Student records, interview records, and communication data shall be protected from unauthorized access.
- Email notifications shall not disclose sensitive message content.
- Only authorized users shall perform Excel imports or Word exports.

5.3.4 Audit and Session Security

- The system shall record login time, logout time, and major user operations for auditing.
- Log information shall be protected from unauthorized access or modification.
- User sessions shall be invalidated after logout to prevent unauthorized reuse.

5.4 Software Quality Attributes

The Mentor Caring System shall emphasize adaptability, reliability, usability, maintainability, correctness, and interoperability to support multiple user roles, yearly mentor

allocation, and future organizational expansion.

5.4.1 Adaptability

- The system shall support the addition of new faculties, departments, and majors without requiring program code modification.
- Organizational structure data shall be configurable manually or via Excel import.
- Annual mentor allocation updates shall preserve previously stored student information.

5.4.2 Reliability

- Student, mentoring, and appointment records shall remain accurate across academic years.
- Previously stored student information shall remain accessible after new allocations are imported.
- The system shall prevent silent data loss during import, export, deletion, or update operations.
- The system shall achieve at least 99% availability during normal semester operation, excluding scheduled maintenance.

5.4.3 Usability

- Users shall only see functions appropriate to their role.
- The system shall provide clear warning and error messages.
- Common tasks such as searching records, sending messages, making appointments, and exporting records shall be completable within three steps after entering the function page.
- The interface shall maintain consistent labels and navigation structures.

5.4.4 Maintainability

- The system shall use a modular design separating user management, messaging, appointment management, import/export, and logging.
- Configuration data such as roles and faculty structures shall be separated from core logic.
- System logs shall support debugging, auditing, and maintenance.

5.4.5 Correctness

- Existing information shall remain unchanged when duplicated data appear in imported Excel files.
- Access restrictions shall be correctly enforced according to user roles and mentoring groups.
- Exported records shall correctly match the selected academic year, department, mentor, and student conditions.

5.4.6 Interoperability

- The system shall integrate with the school account database for authentication and role identification.
- The system shall support Excel import and Word export according to the required formats.
- The system shall interoperate with an email service to send message notifications.

5.4.7 Quality Attribute Priority

- Security and correctness shall have the highest priority.
- Reliability shall follow security and correctness.
- Usability shall ensure effective system usage by different roles.
- Adaptability and maintainability shall support future system expansion and evolution.

6. Other Requirements

6.1 Database requirements

The system shall connect to the school database to manage user accounts and project group information. The database should support role-based access control to ensure that different users have appropriate permissions within the system.

All previous care records of students must be retained. Even if a student is removed from the current group or changes mentors, the relevant data should remain accessible.

The school database connected with the system should contain the following information:

- **User accounts:** for staff and students
- **Passwords:** used for user authentication
- **Names:** the name of the users
- **User Roles**
- **Faculty information**
- **Department information**
- **Major information**
- **Student IDs**
- **Group IDs**
- **Mentor information:** associated with each group, including mentor name and mentor email
- **Academic year information:** associated with group allocation
- **MCP coordinator information:** including coordinator name, email, and department
- **Student mentoring records and history**

The system should support multiple user roles, including:

- Administrator (usually only one)

- MCP Coordinator
- Faculty Consultants
- Mentors
- Students
- Supporting Staff

Each role should have different permissions and access levels within the system.

The database should ensure the following:

- Secure storage of user credentials
- Controlled access to sensitive data
- Efficient retrieval of user information for login and authorization purposes
- Support for importing and updating data from Excel templates
- Consistency between imported group allocation data and existing student records

6.2 Internationalization requirements

The system shall support internationalization to accommodate users from different linguistic backgrounds. English should be provided as the default language of the system interface. The system design should allow future extension to support additional languages if required.

6.3 Legal requirements

The system must comply with relevant institutional regulations regarding data privacy and information protection. Personal information of users, including student records and mentor information, must be protected and accessible only to authorized users.

6.4 Project reuse objective

The system should be designed with modular components so that certain modules (such as the user authentication module and database access module) can be reused in future university software systems if needed.

Appendix A: Glossary

MCP: Mentor Caring Program.

Faculty Consultant: Departmental advisor, responsible for designating the MCP coordinator and the import list.

MCP Coordinator: Departmental Coordinator, responsible for handling special cases forwarded by mentors.

Appendix B: Analysis Models

Context model in Section 2.1

Use case diagram in Section 2.2

Appendix C: Issues List

The following issues remain to be clarified during later stages of the system development process:

- Final confirmation of database schema
- Detailed access permissions for each user role
- Possible additional system features for future versions
- Final design of the user interface

These issues will be reviewed and resolved during the system design phase.